EMS611U: Conceptual Design of Aircraft

Dr. Hülya Biler
Queen Mary University of London

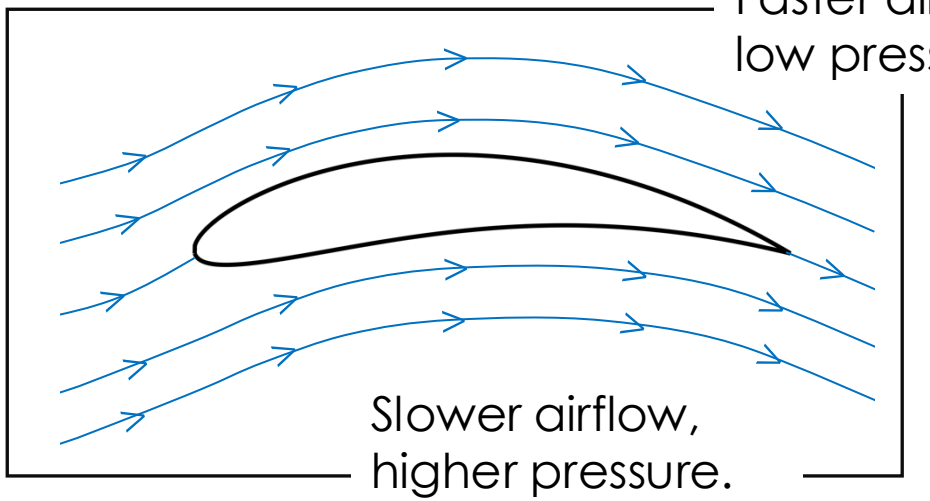
Week 6

Wing Theory

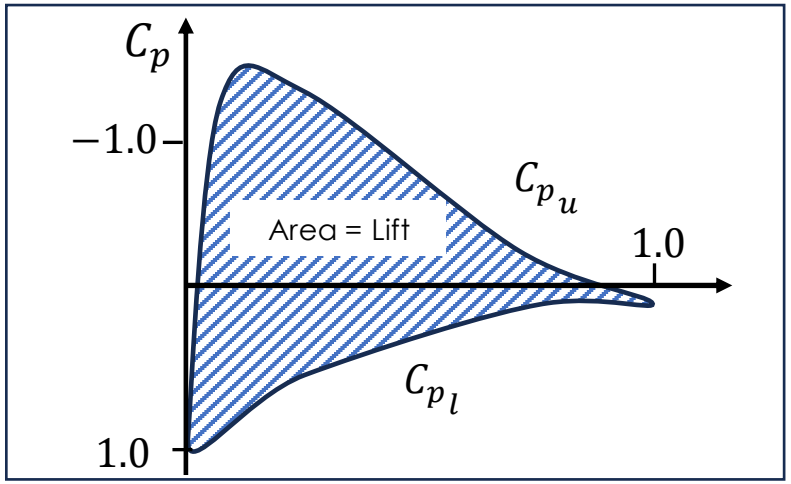
Road Map

We have focused on **aerofoil** so far.

Flow Around an Aerofoil



Pressure and Lift

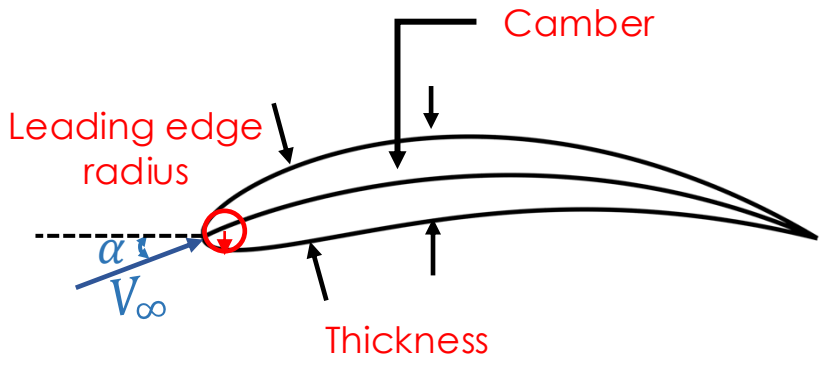


Road Map

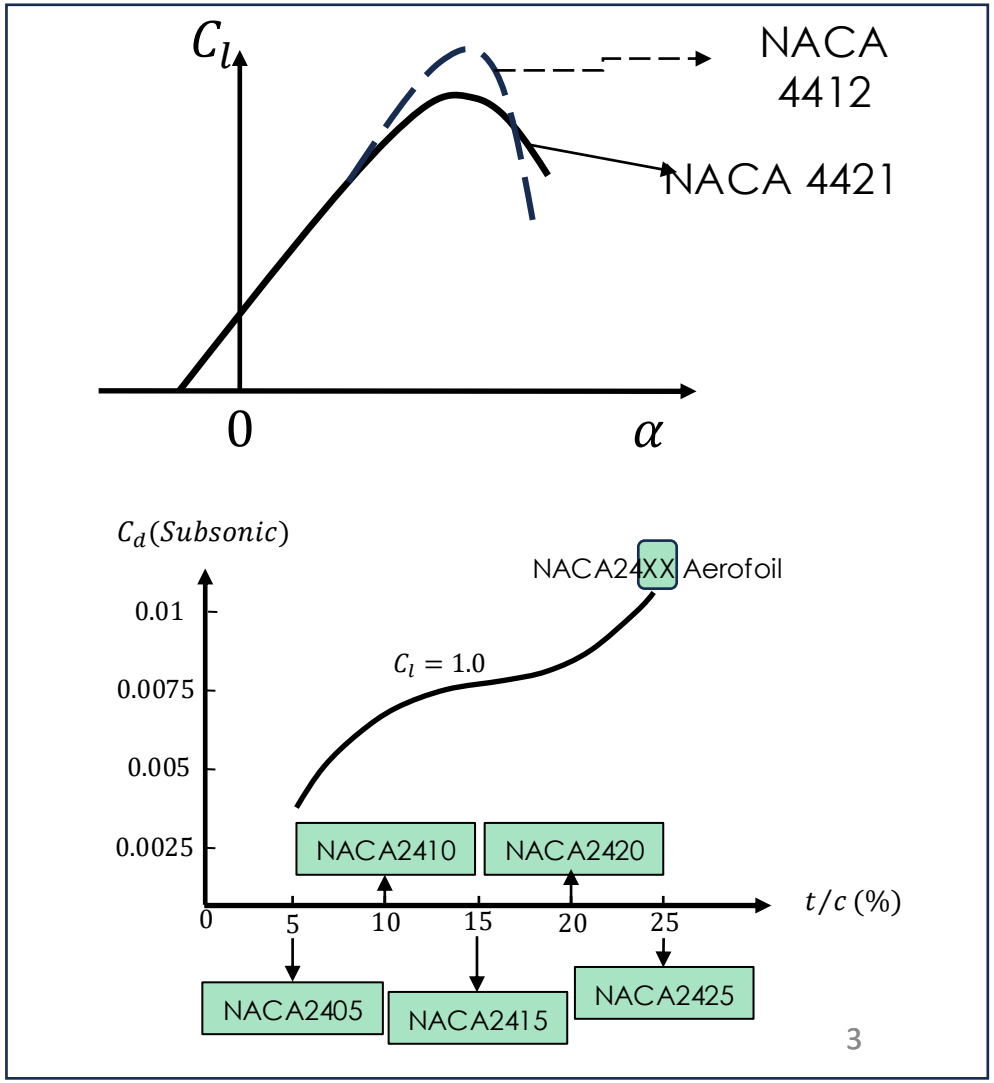
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Flow Around an Aerofoil and its Implication on Pressure and Lift

Geometric Factors and Their Effects



Aerofoil Families & Stall, Lift and Drag Chacteristics



Road Map

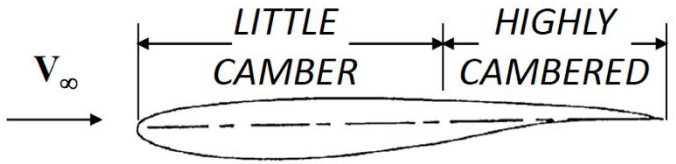
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Flow Around an Aerofoil and its Implication on Pressure and Lift

Geometric Factors , Aerofoil Families and Their Effects on stall, lift and drag characteristics

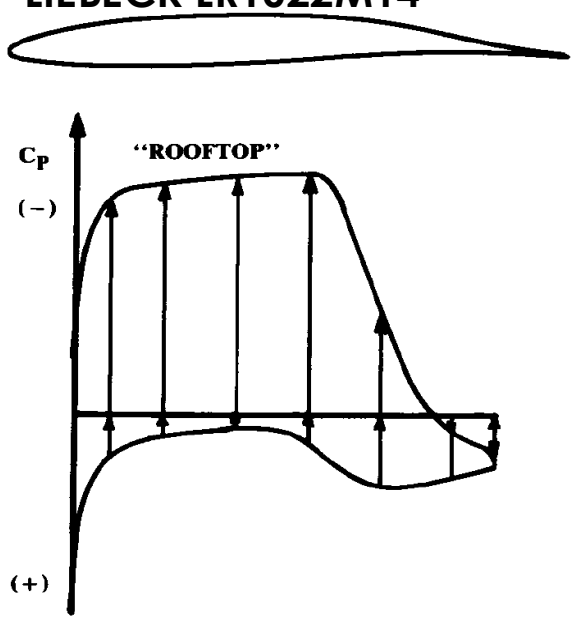
Aerofoil Design to Serve a Specific Purpose

Supercritical Aerofoils



Laminar Aerofoils

LIEBECK LR1022M14



Road Map

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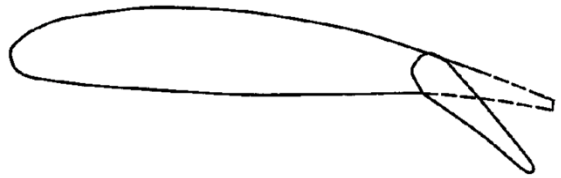
Flow Around an Aerofoil and its Implication on Pressure and Lift

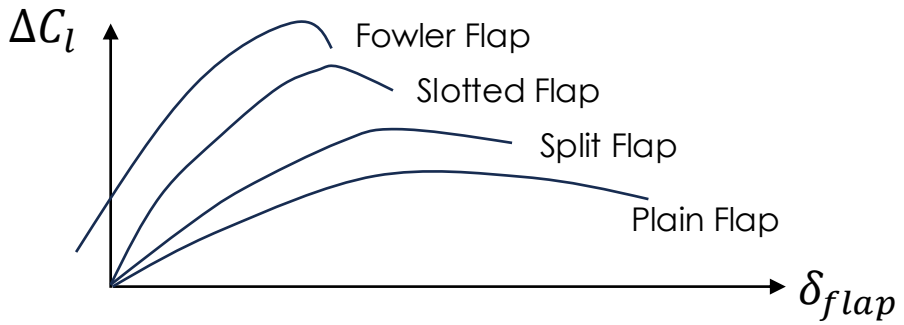
Geometric Factors , Aerofoil Families and Their Effects on stall, lift and drag characteristics

Aerofoil Design to Serve a Specific Purpose

Effect of High Lift Devices

- Trailing Edge Flaps
- Leading Edge Devices
- Boundary Layer Control





Road Map

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Effect of High Lift Devices

Geometric Factors , Aerofoil Families and Their Effects on stall, lift and drag characteristics

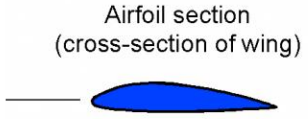
Aerofoil Characteristics Two-dimensional

C_l

C_d

C_m

Aerofoil Design to Serve a Specific Purpose



Road Map

We have focused on **aerofoil** so far.

Flow Around an Aerofoil and its Implication on Pressure and Lift

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Aerofoil Design to Serve a Specific Purpose

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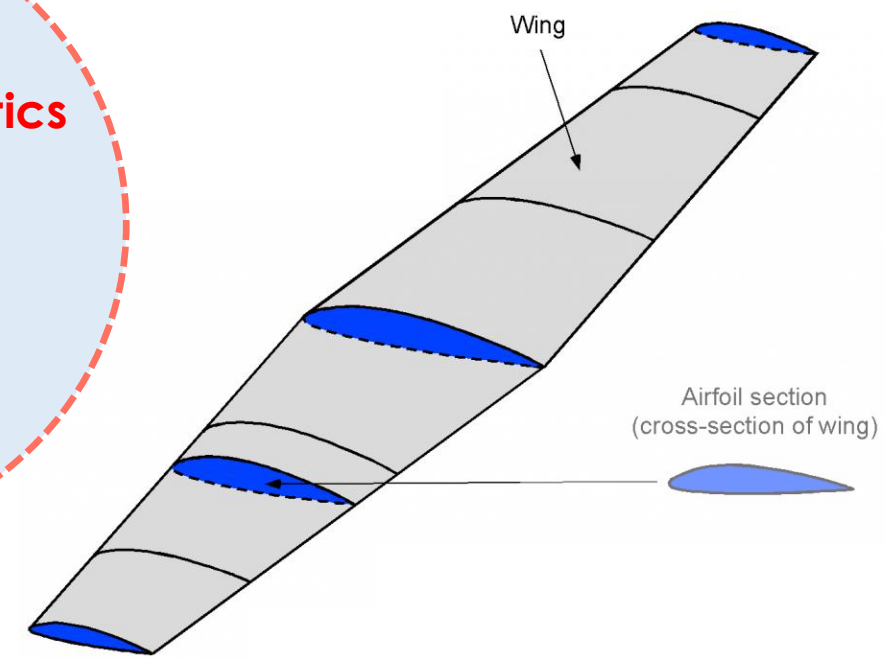
C_m

Now we will focus on: Wing Characteristics Three-dimensional

C_L

C_D

C_M



Wing Aerodynamics



Beechcraft Baron 58

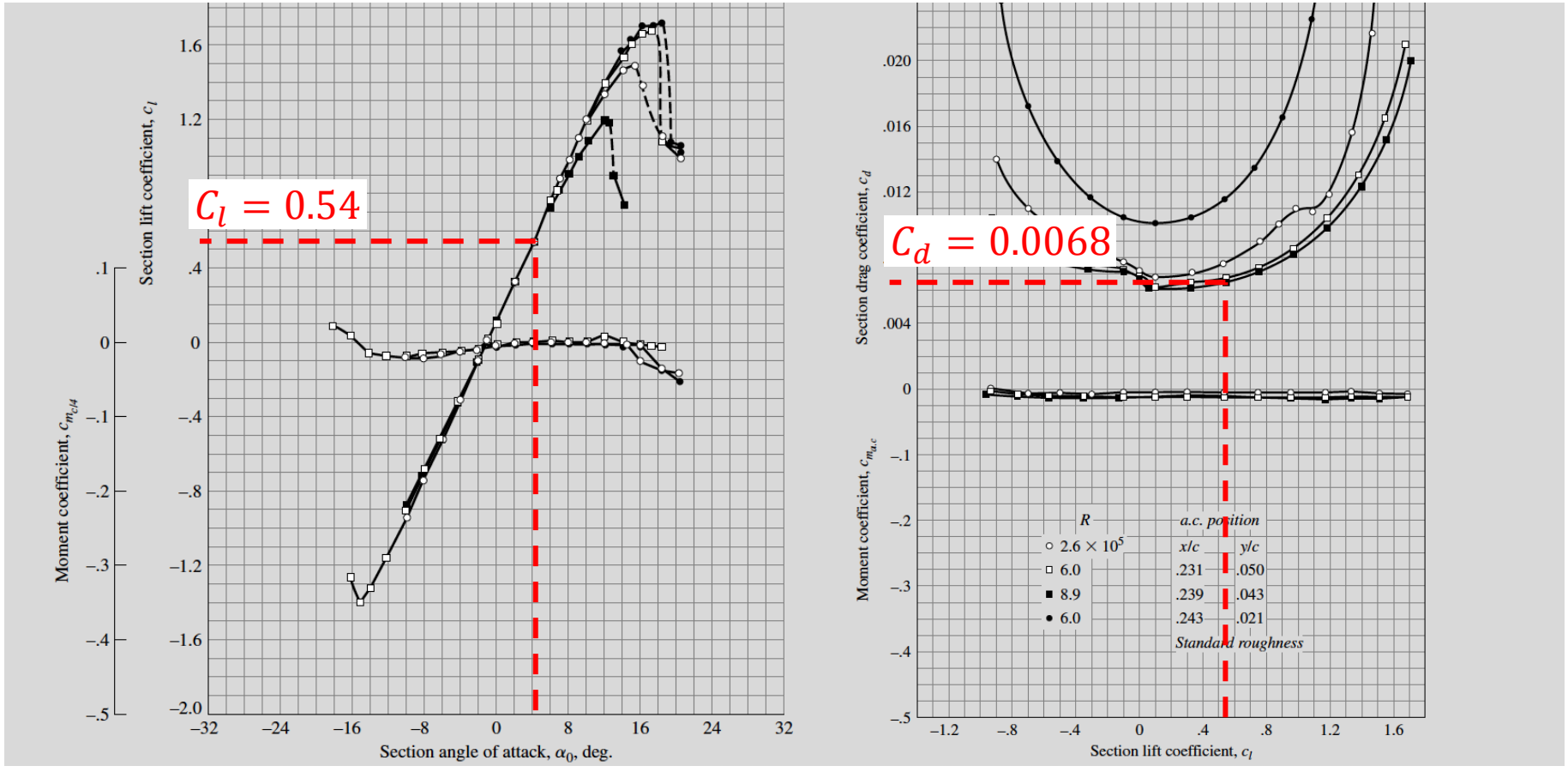
Let's look at:

- Beechcraft Baron 58 twin-engine business aircraft
- The wing on this airplane has a 15 percent thick **NACA 23015** aerofoil at the root, tapering to a 10 percent thickness at the tip.

Beechcraft Baron 58

Could we just look at the NACA23015 aerofoil data and determine the lift and drag coefficients? **NO!**

- Let's see what the aerofoil data show for $\alpha = 4°, Re = 6\ x\ 10^5$
  - $C_l = 0.54$
  - $C_d = 0.0068$

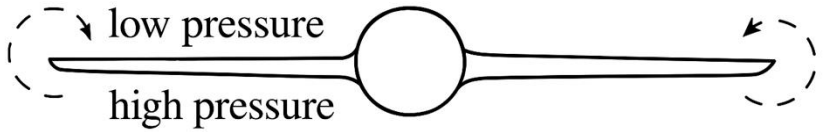


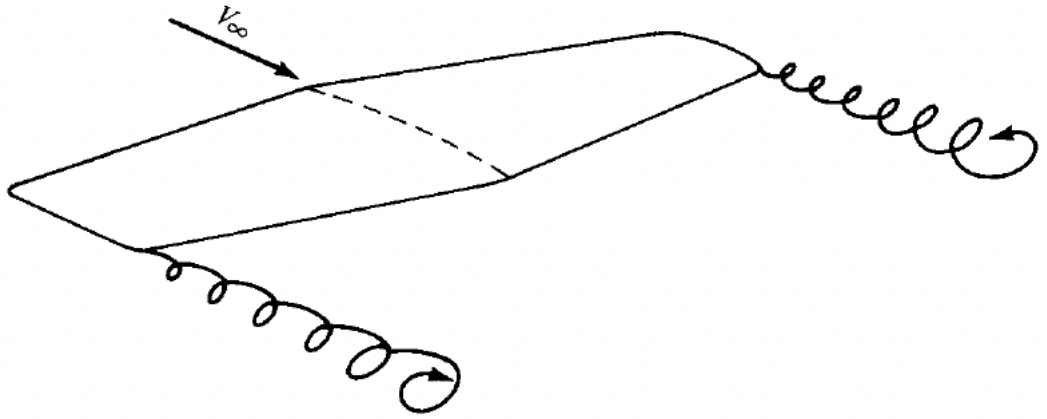
These values are not same as the wing C_L and C_D

Why they are not the same? and How can we correct for the differences?

Wing Aerodynamics Recap

- Wing Aerofoils are the same as the properties of a wing of infinite span; indeed, aerofoil data are frequently denoted as "infinite wing" data. However, all real airplanes have wings of finite span.
- Aerodynamics is three-dimensional.
  - **There is a spanwise velocity contribution.**





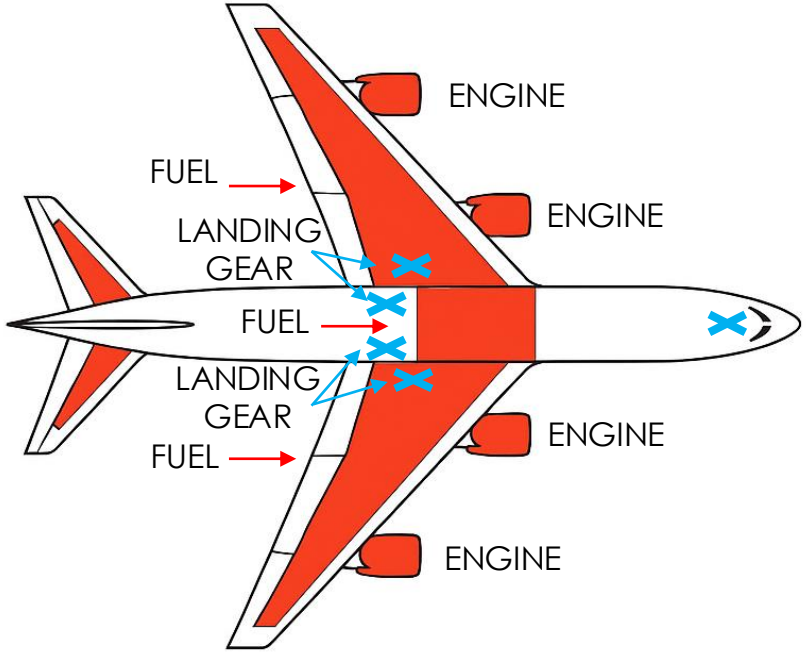
- When a wing is generating lift, it has a reduced pressure on the upper surface and an increased pressure on the lower surface.

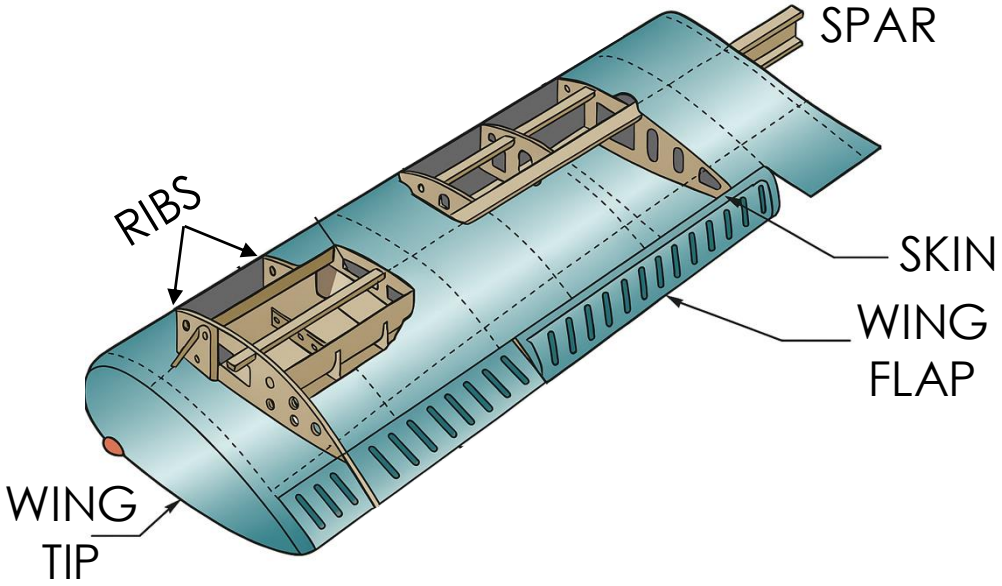
- Near the tip of the wing, the high-pressure air will slip around to reach the top of the wing. This circulation of air around the tip creates a vortex and also pushes down on the top of the wing, **spoiling lift and creating drag.**

Wing Geometry

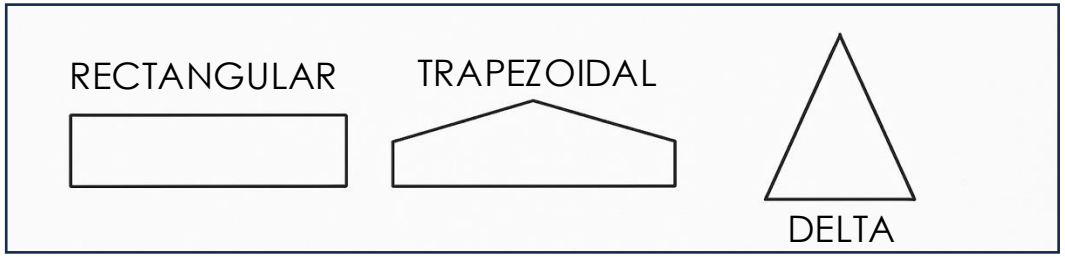
Important considerations/constraints:

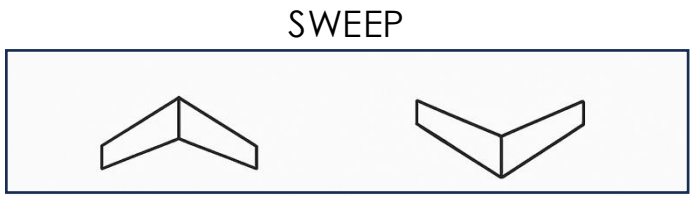
- Performance (cruise, loiter, take-off, landing)
- Flying qualities (handling and stability)
- Structural considerations (spar placement)
- Internal volume (for fuel/payload)
- Stealth characteristics (for military subsonic)
- Airport limitations (wing-span)

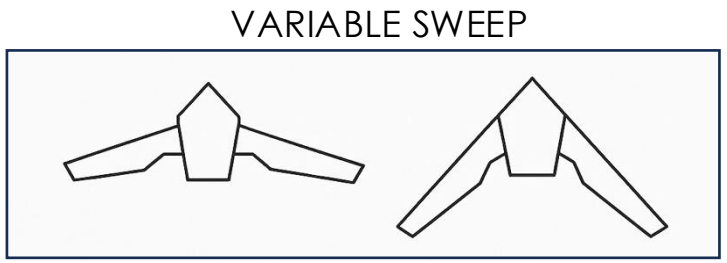




WING PLANFORM





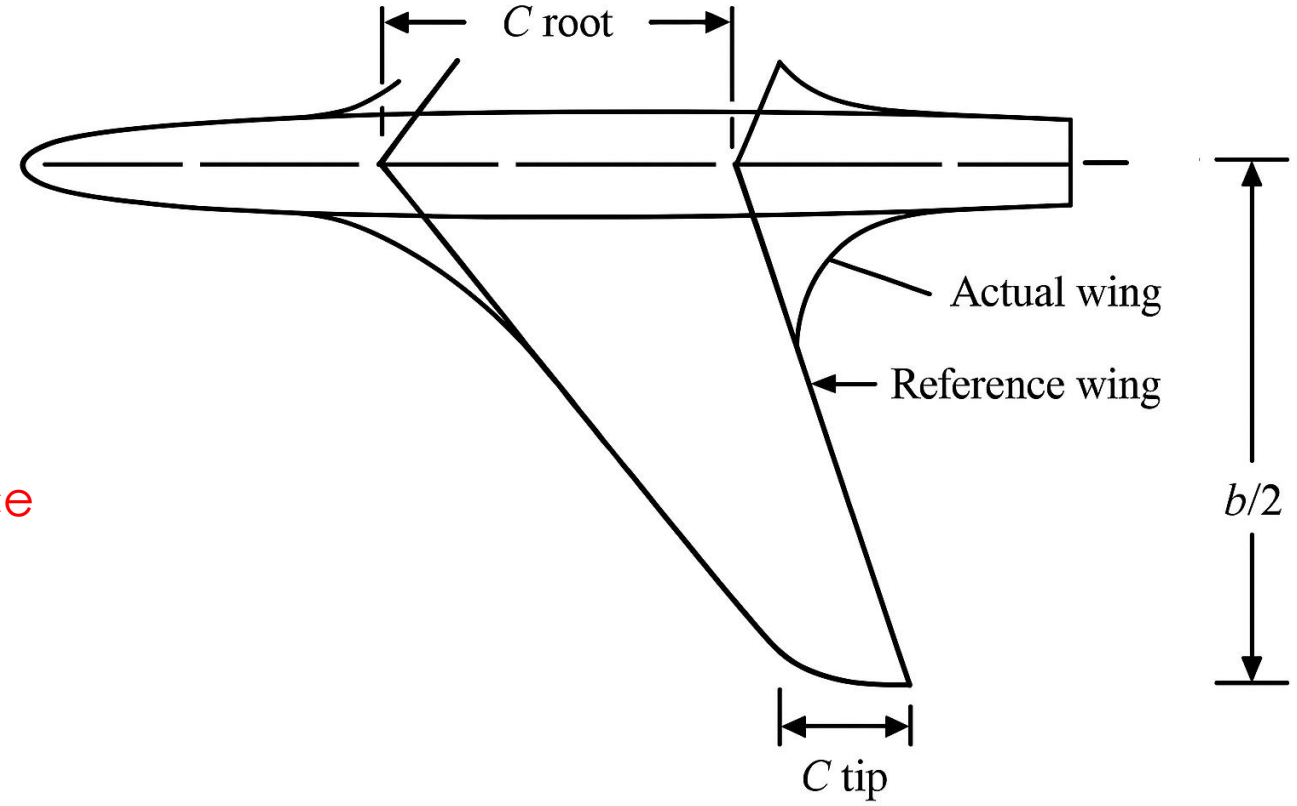


Wing Geometry

Planform area:
It is the area of wing planform obtained by extending the exposed wing up to the fuselage centre line.

Reference (or Equivalent) wing:
It is a trapezoidal wing whose root chord is at the fuselage centre line and has the area same as the planform area.

The root aerofoil is the aerofoil of the trapezoidal reference wing at the centerline of the aircraft, not where the actual wing connects to the fuselage.



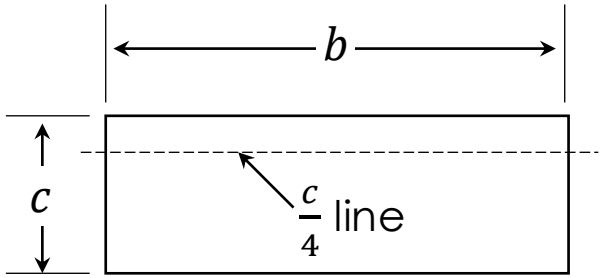
The reference wing area is fictitious:

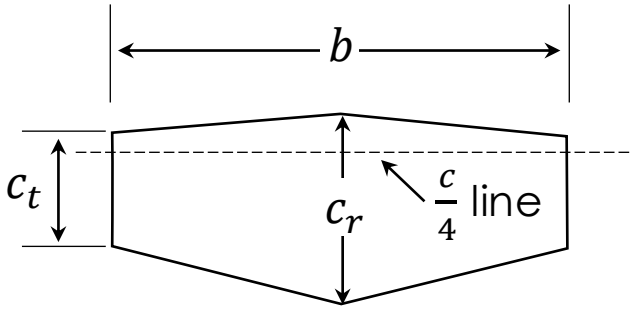
- Wing area includes the part of the reference wing that sticks into the fuselage.

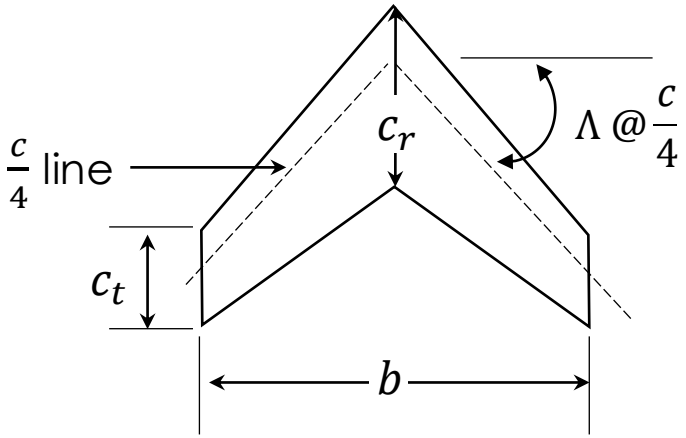
Wing Geometry

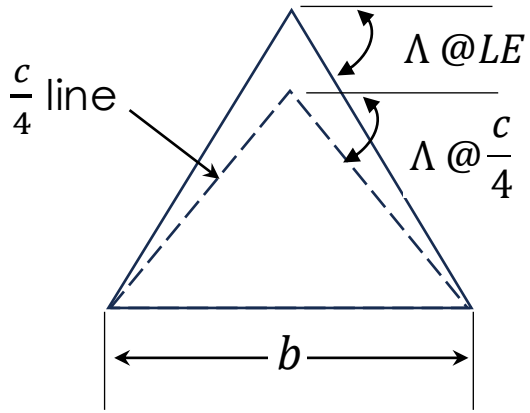
The "reference" wing is the basic wing geometry used to begin the layout:

- S, reference wing area
- c, chord
- b, span
- A, aspect ratio (b^2/c)
- t/c, thickness ratio
- Λ, sweep
- λ, taper ratio (c_{tip}/c_{root})
- Twist
- Dihedral









Aspect Ratio

- Early wings were rectangular, so aspect ratio was defined as span divided by chord.
- For tapered wings, it's span squared divided by area.

For rectangular wings

$$A = \frac{b}{\bar{c}}$$

For delta wings

$$A = \frac{b^2}{S}$$

S, reference wing area
$\bar{c}$, mean chord
b, span

Aspect ratio affects:

- The lift and slope of the lift curve of the wing (C_{L_α}).
- Induced drag (C_{D_i}).
- The structural weight of the wing.
- The wing span.

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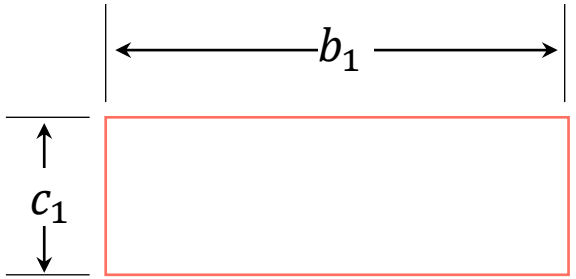
Aspect Ratio

- Effect of aspect ratio on slope of the lift curve:

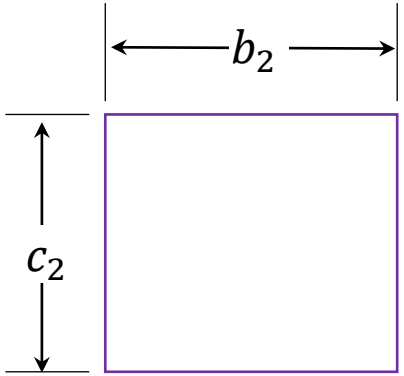
- The Wright Brothers discovered that a long, narrow wing (high aspect ratio) produces higher lift compared to a short, wide wing (low aspect ratio) via wind tunnel testing.

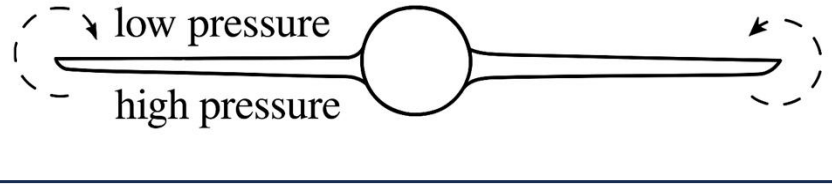
due to 3D effects.

Wing 1



Wing 2





This air escape causes a lift reduction near the wing tips.

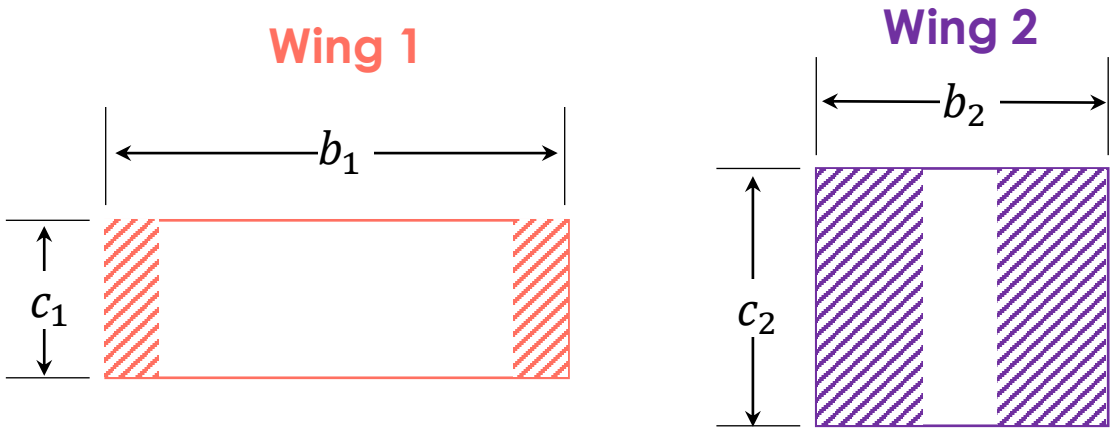
- $S_1 = S_2$, if the wing areas of the two wings are the same.
- We have very different aspect ratios for these two wings, $A_1 > A_2$

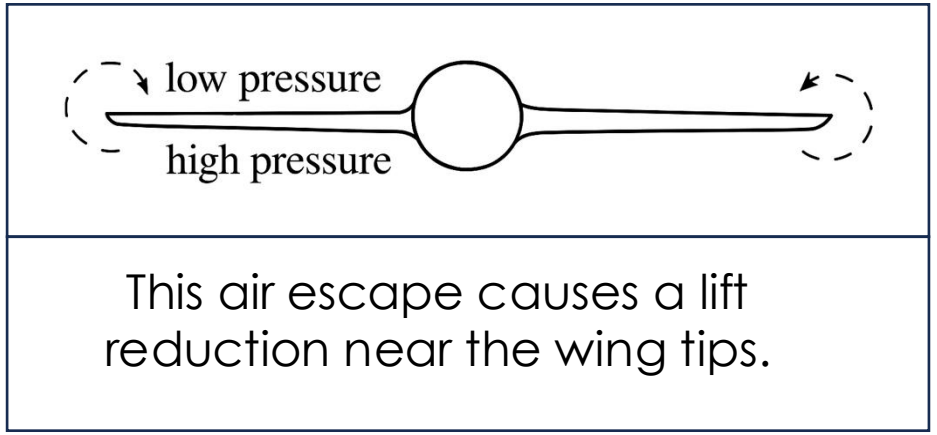
Aspect Ratio

- Effect of aspect ratio on slope of the lift curve:

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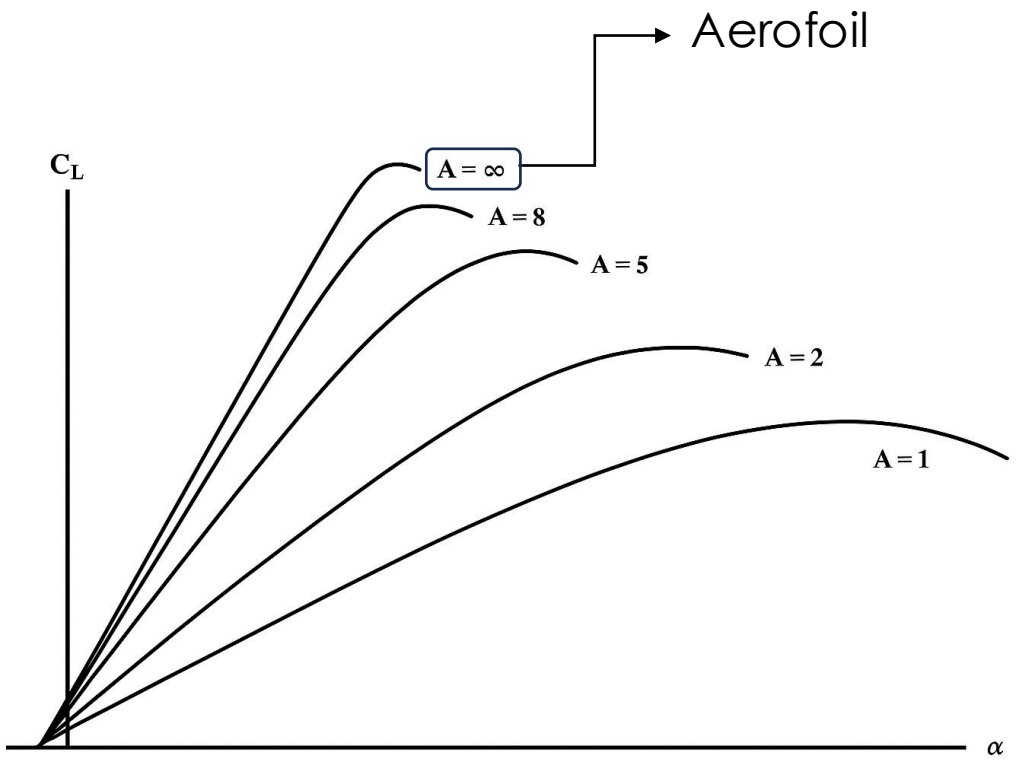
- $S_1 = S_2$, if the wing areas of the two wings are the same.
- We have very different aspect ratios for these two wings, $A_1 > A_2$

Notes:

- The amount of wing affected by the tip vortex is more for the lower aspect ratio wing.
- Therefore, the high aspect ratio wing does not experience as much of a lift loss as the lower aspect ratio wing.

Aspect Ratio

- Effect of aspect ratio on slope of the lift curve:

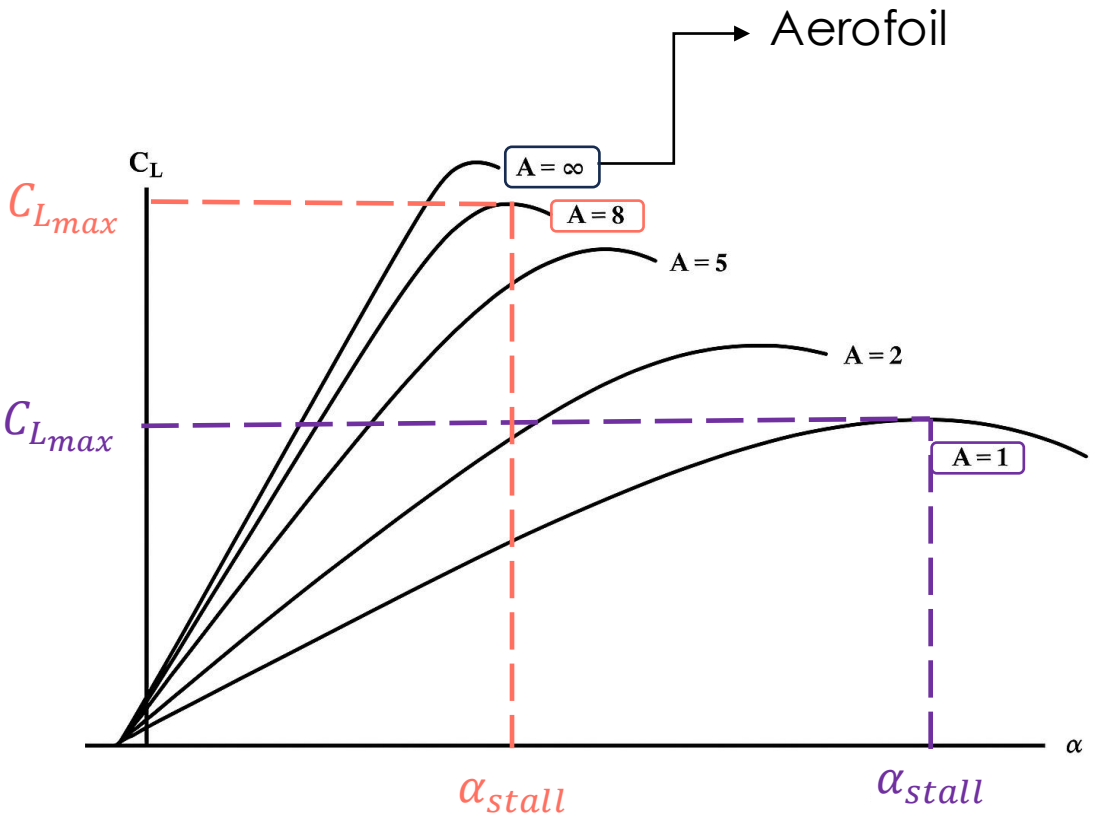


Notes:

- Larger the aspect ratio, larger the lift coefficient

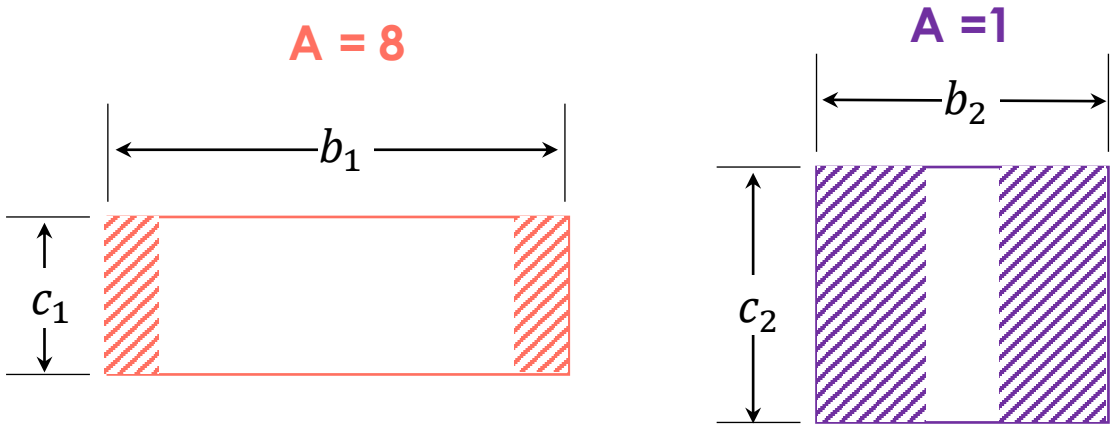
Aspect Ratio

- Effect of aspect ratio on slope of the lift curve:



Notes:

- Larger the aspect ratio, larger the lift coefficient

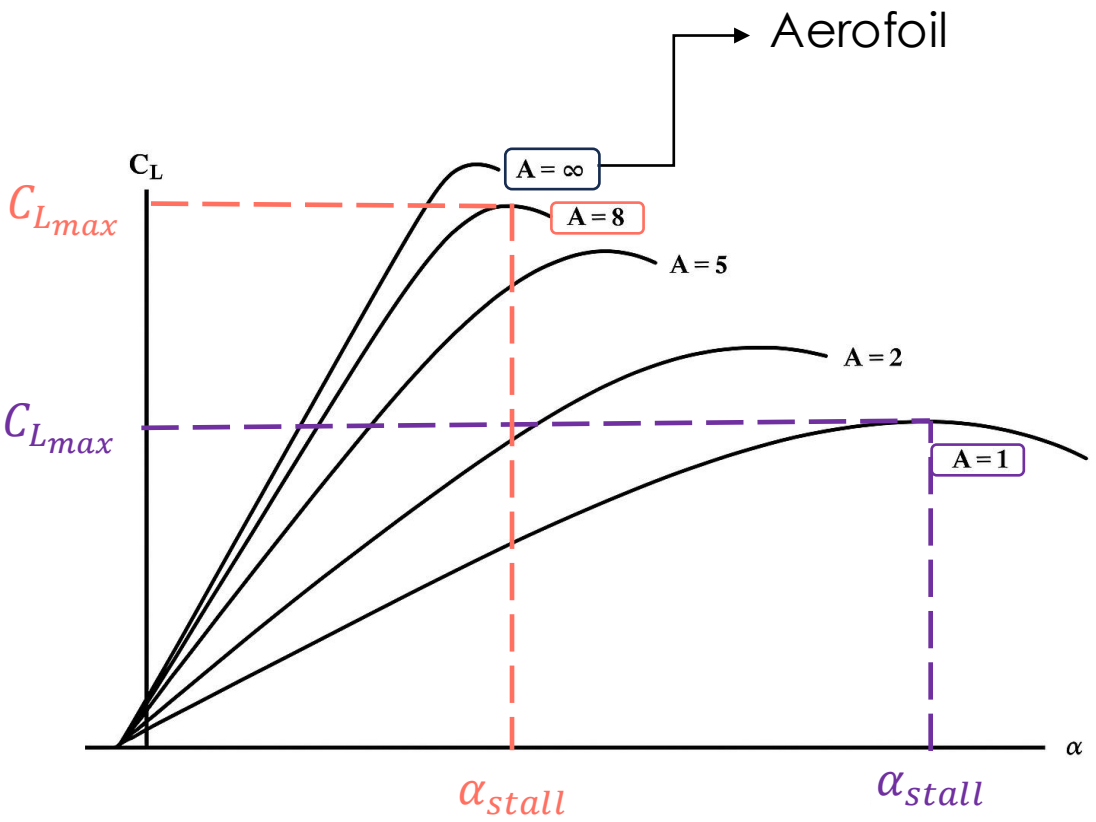


- The reduced effective angle of attack at the tips, a lower aspect ratio wing will stall at a higher geometric angle of attack.

Due to higher stall angle observed for lower aspect ratios, lower aspect ratio wings are generally used for tails. Delaying tail stall until well after the wing stall assures adequate control.

Aspect Ratio

- Effect of aspect ratio on slope of the lift curve:

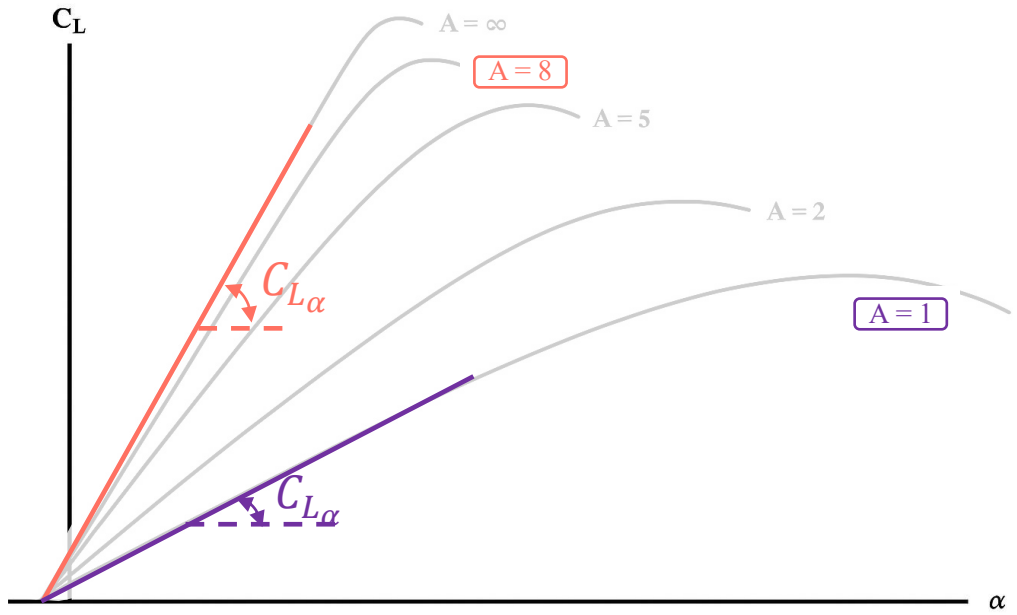


Notes:

- Larger the aspect ratio, larger the lift coefficient
- As the aspect ratio increases stall angle of attack decreases.

Aspect Ratio

- Effect of aspect ratio on slope of the lift curve:



Notes:

- Larger the aspect ratio, larger the lift coefficient
- As the aspect ratio increases stall angle of attack decreases.
- As the aspect ratio increases lift slope curve increases.

Aspect Ratio

- Effect of aspect ratio on induced drag:

- A wing with a high aspect ratio will generate more lift and less induced drag than a wing with a low aspect ratio.

- Effect of aspect ratio on the wing span:

- For a chosen wing area, the aspect ratio determines the span of the wing.
- In turn the span determines the **hanger space** needed for the airplane.

- Effect of aspect ratio on the structural weight of the wing:

$$W_{wing} = CS_W^{0.649} A^{0.5} (t/c)_{root}^{-0.4} (1+\lambda)^{0.1} (\cos\Lambda)^{-1}$$

- Equation shows that the wing weight increases as square root of the aspect ratio.
- The reason for this is that the span increases as the aspect ratio increases.
- An increase in the span would increase the bending moment at the wing root.
- **This would require higher moment of inertia of the spar and hence higher weight.**

Aspect Ratio

- **Trade-offs:**

- Larger the aspect ratio, larger the lift coefficient.
  - However, **for a given wing area,** increasing the aspect ratio may result in a **too small wing chord** with a **too low Reynolds number**, which may significantly **reduce the lift coefficient**.

$$Re = \frac{\rho V_\infty c}{\mu_\infty}$$

  - It requires **higher moment of inertia of the spar** and hence **higher weight**.

Aspect Ratio

- In this design stage, the aspect ratio will be determined by a trade study in which the aerodynamic advantages of a higher aspect ratio are balanced against the increased weight etc.

For initial wing layout, the values and equations provided in the table can be used.

Propeller Aircraft	Aspect Ratio
Flying boat	8.0
Twin turboprop	9.2
Agriculture aircraft	7.5
General aviation-twin engine	7.8
General aviation-single engine	7.6
Homebuilt	6.0

Jet Aircraft	a	c
Jet transport	7.5	0
Military cargo/bomber	5.57	-1.075
Jet fighter (other)	4.11	-0.622
Jet fighter (dogfighter)	5.416	-0.622
Jet trainer	4.737	-0.979

$$A = aM_{max}^{c}$$

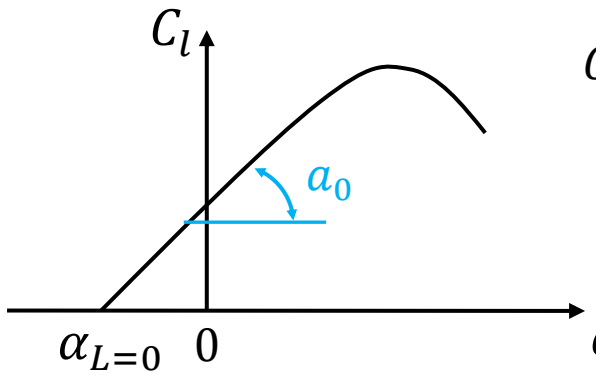
M_{max}: Maximum flight Mach Number

- Jet aircraft show a strong trend of aspect ratio decreasing with increasing Mach number.
- This is probably due to drag-due-lift becoming relatively less important at higher speeds.
- Designers of high-speed aircraft thus use lower-aspect-ratio wings to save weight.

Aspect Ratio Correction

Reminder

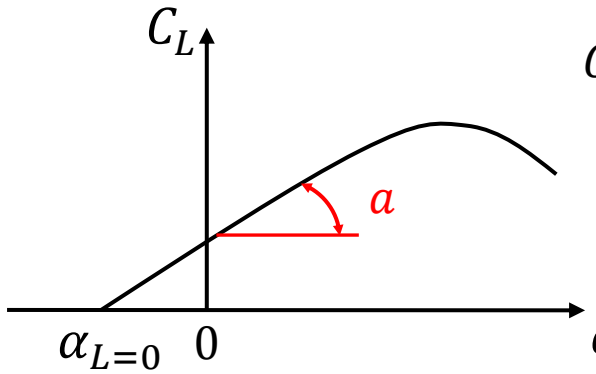
2D Lift coefficient



$$C_l = a_o(\alpha - \alpha_{L=0})$$

$$a_0 = \frac{dC_l}{d\alpha}$$ 2D lift curve slope

3D Lift coefficient



$$C_l = a(\alpha - \alpha_{L=0})$$

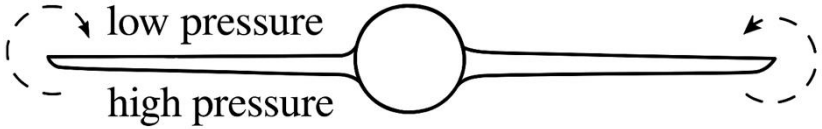
$$a = \frac{dC_L}{d\alpha}$$ 3D lift curve slope

α: Geometric Angle of attack

$\alpha_{L=0}$: Zero-lift angle of attack

How can we relate 2D lift curve slope (a_0) and 3D lift curve slope (a) ?

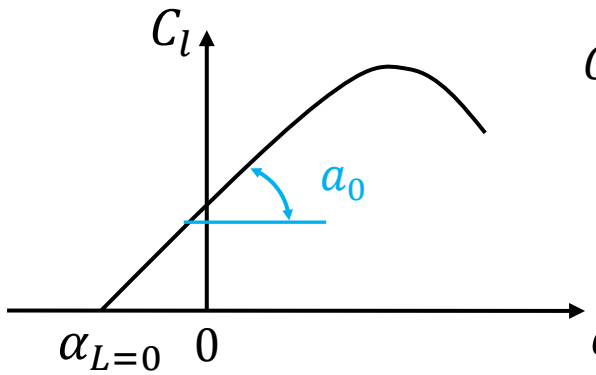
What do we know about 2D and 3D flows?



Aspect Ratio Correction

Reminder

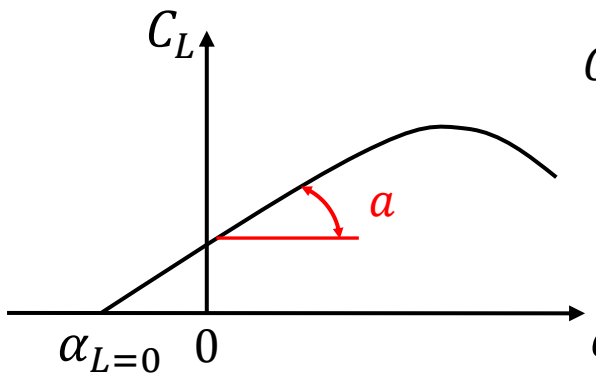
2D Lift coefficient



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$$a_0 = \frac{dC_l}{d\alpha}$$ 2D lift curve slope

3D Lift coefficient



$$C_l = a(\alpha - \alpha_{L=0})$$

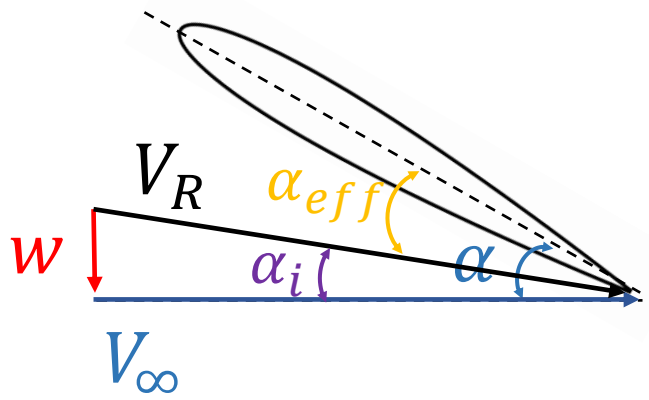
$$a = \frac{dC_L}{d\alpha}$$ 3D lift curve slope

α: Geometric Angle of attack

$\alpha_{L=0}$: Zero-lift angle of attack

How can we relate 2D lift curve slope (a_0) and 3D lift curve slope (a) ?

What do we know about 2D and 3D flows?



α_i : Induced angle of attack

α_{eff} : Effective angle of attack, $\alpha_{eff} = \alpha - \alpha_i$

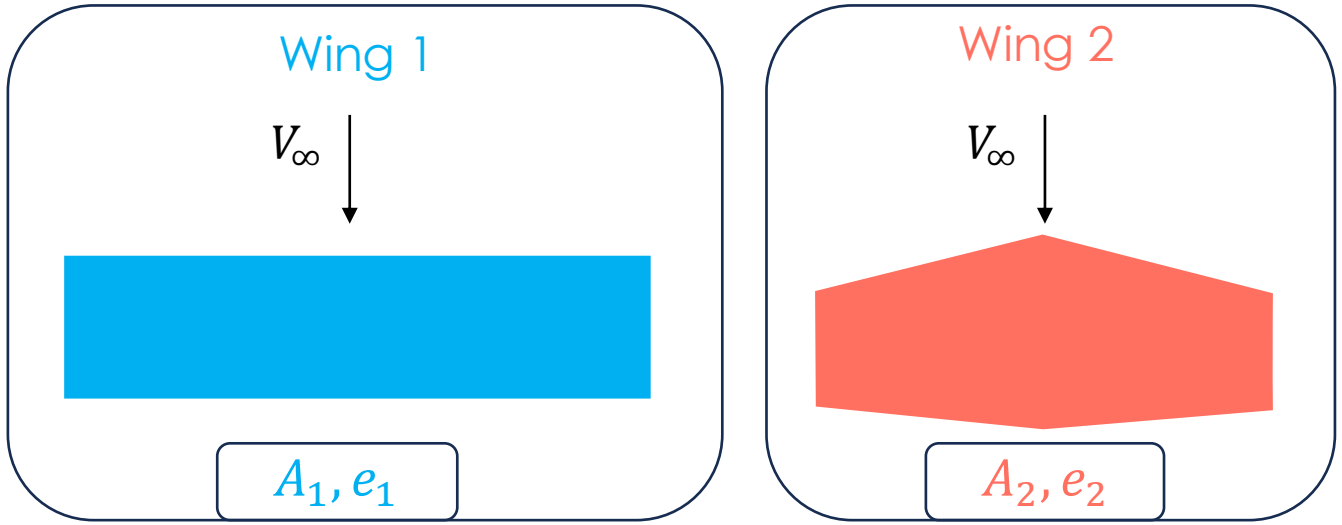
- In 3D → α_i (Due to downwash) !!
- Different wings create different induced angles of attack.

$$\alpha_i = \frac{C_L}{\pi A e}$$

$A = \frac{b^2}{S_w}$, Aspect ratio

e, Oswald's efficiency factor

Aspect Ratio Correction



Let's say that both wings have the same aerofoil.

If they are at the same effective angle of attack (essentially taking out the induced angle of attack, 3D effects), these two wings should create same lift coefficients.

$$\alpha_{eff_1} = \alpha_{eff_2}$$

$$\alpha_{eff} = \alpha - \alpha_i$$

$$\alpha_1 - \alpha_{i_1} = \alpha_2 - \alpha_{i_2}$$

$$\alpha_i = \frac{C_L}{\pi A e}$$

$$\alpha_1 - \frac{C_L}{\pi A_1 e_1} = \alpha_2 - \frac{C_L}{\pi A_2 e_2}$$

$$C_L = a(\alpha - \alpha_{L=0})$$

$$\frac{C_L}{a} + \alpha_{L=0} = \alpha$$

$$\frac{C_L}{a_1} + \alpha_{L=0} - \frac{C_L}{\pi A_1 e_1} = \frac{C_L}{a_2} + \alpha_{L=0} - \frac{C_L}{\pi A_2 e_2}$$

$$\frac{C_L}{a_1} = \frac{C_L}{a_2} + \frac{C_L}{\pi}\left(\frac{1}{A_1 e_1} - \frac{1}{A_2 e_2}\right)$$

$$a_1 = \frac{a_2}{1 + \frac{a_2}{\pi}\left(\frac{1}{A_1 e_1} - \frac{1}{A_2 e_2}\right)}$$

Aspect Ratio Correction

$$a_1 = \frac{a_2}{1 + \frac{a_2}{\pi}\left(\frac{1}{A_1 e_1} - \frac{1}{A_2 e_2}\right)}$$

$$a_1, a_2, a_0 \; in \left[\frac{1}{rad}\right]$$

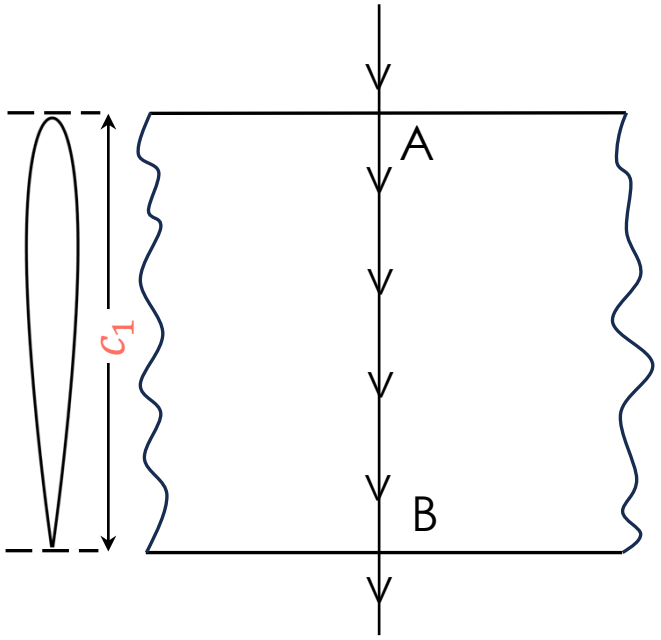
Let's say Wing 2 is an infinitely long wing, $A_2 = \infty$,
(equivalent to an aerofoil), a_0.

$$a_1 = \frac{a_0}{1 + \frac{a_o}{\pi A e}}$$

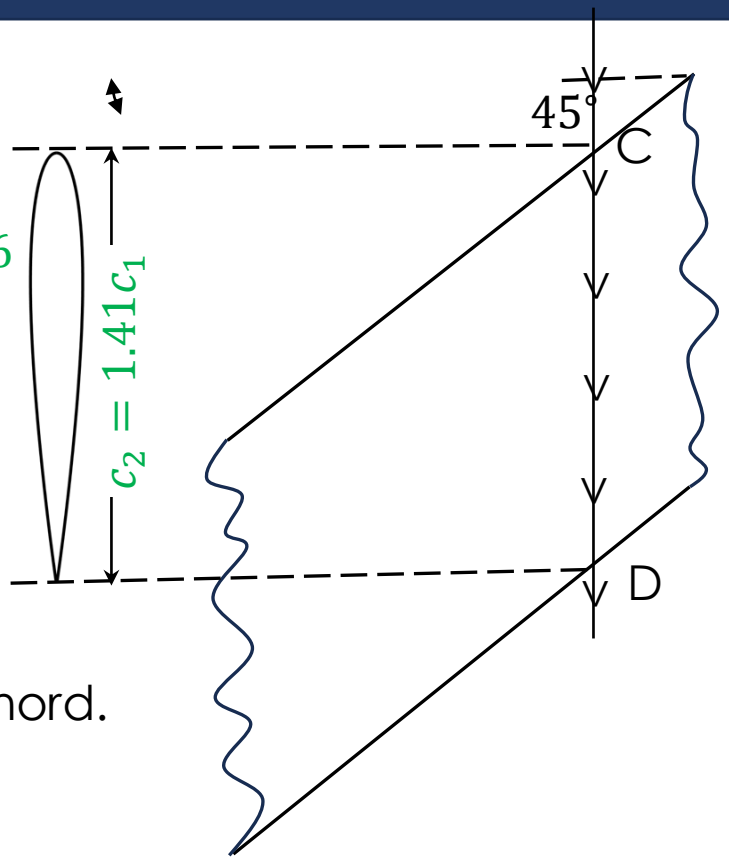
We relate 2D lift curve slope (a_0) and 3D lift curve slope (a) !!

Wing Sweep

$\frac{t_1}{c_1} = 0.15$



$\frac{t_2}{c_2} = 0.106$



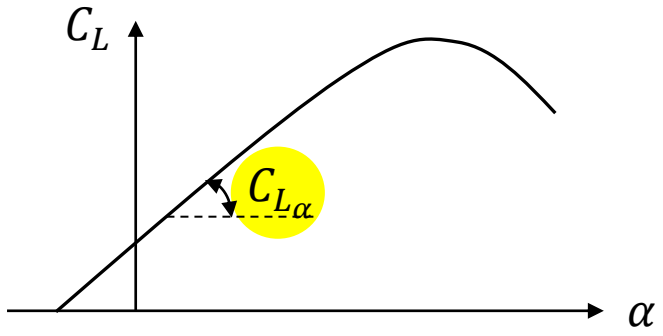
- Aerofoil has same thickness, but no longer the same effective chord.
- Effective aerofoil section is thinner,
- Making aerofoil thinner increases critical Mach Number.

The wing sweep affects:

- **Slope of the lift curve**
- **Induced drag coefficient**
- **Critical Mach Number**
- **Wing Weight**

Wing Sweep

- Effect of sweep on slope of lift curve:



$$C_{L_\alpha} = \frac{2\pi A}{2 + \sqrt{4 + \frac{A^2\beta^2}{\eta^2} + \left(1 + \frac{\tan^2 \Lambda_{max,t}}{\beta^2}\right)}}$$

$$\beta^2 = 1 - M^2$$

$$\eta^2 = \frac{C_{l_\alpha}}{2\pi/\beta}$$

- $\Lambda_{max,t}$: sweep of the line of maximum thickness
- C_{l_α}: slope of the aerofoil lift curve for a given Mach Number

- C_{L_α} decreases as sweep increases.

For instance:
For $A = 8$ wing, where $M = 0.8$.

If we increase the sweep from 0° to 35°

C_{L_α} decrease about 25%.

Wing Sweep

- Effect of sweep on induced drag:

Based on experimental data on swept wing, induced drag of a swept wing is inversely proportional to cosine of $\Lambda - 5°$.

$$C_{D_i} \propto \frac{1}{\cos(\Lambda - 5°)}$$

Wing Sweep

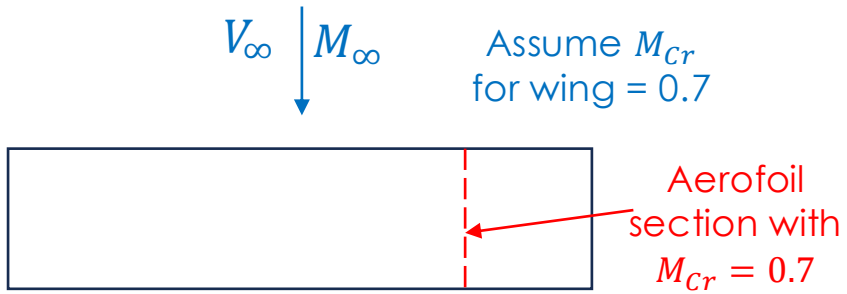
- Effect of sweep on Mach critical (M_{crit}) or Mach drag-divergence number (M_{DD}):

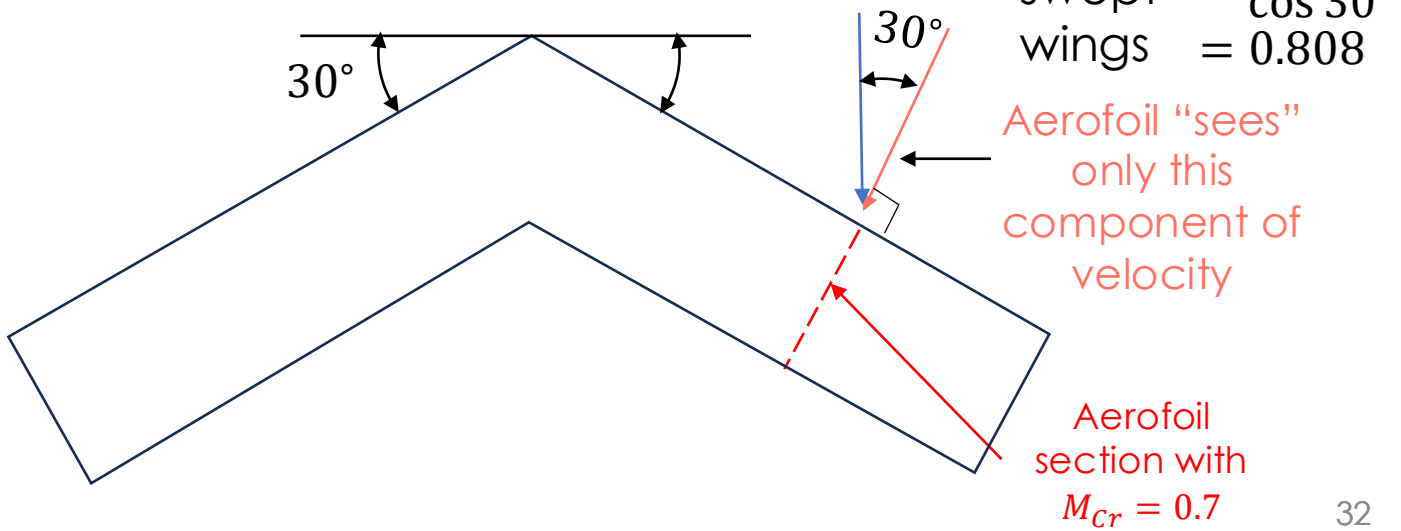
Reminder

- *The critical Mach number in connection with the aerofoil was defined as the free stream Mach number at which the maximum Mach number on the aerofoil is unity.*
- However, when the critical Mach number is exceeded, the drag coefficient starts to increase.

- *Making use of this behaviour we define the term "Drag divergence Mach Number (M_{DD})" as the Mach number at which the drag coefficient shows an increase of 0.002 over the subsonic drag value.*

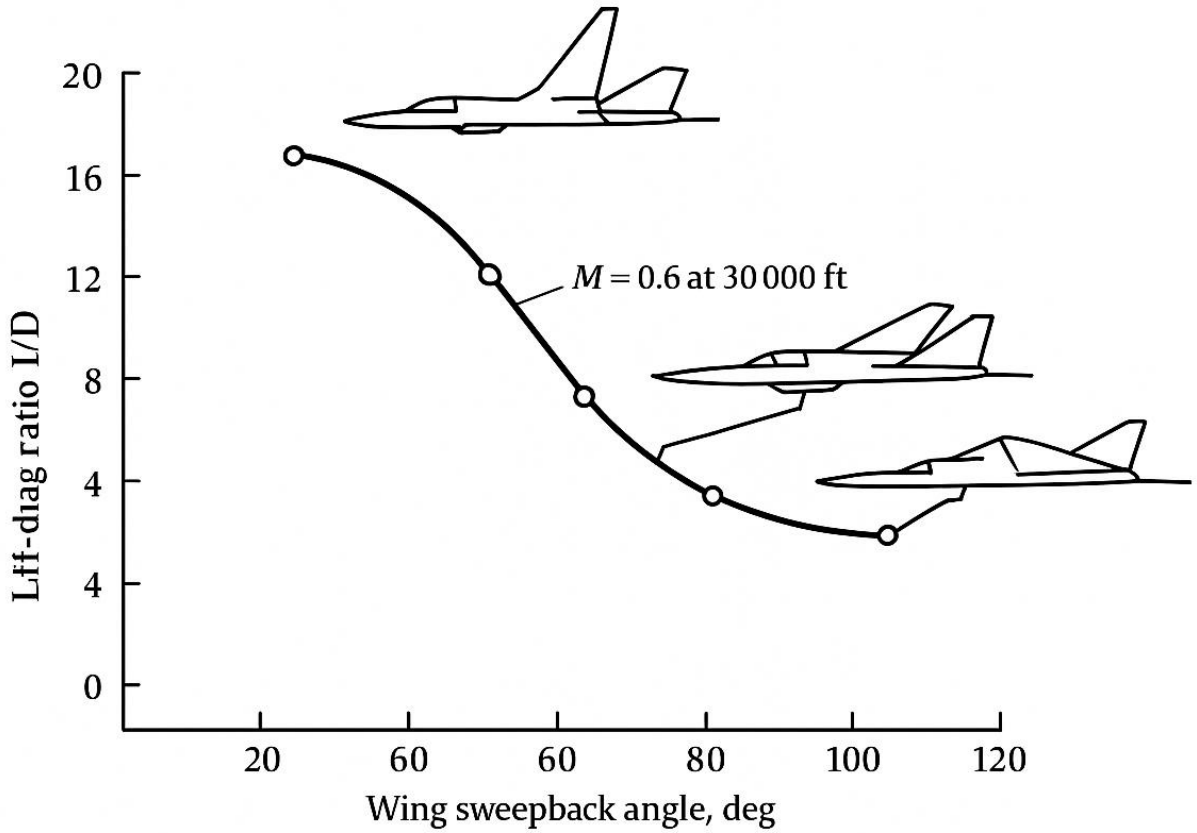
M_{cr} for swept wings $\approx \frac{0.7}{\cos 30^\circ} = 0.808$





Wing Sweep

- Effect of sweep on Mach critical (M_{crit}) or Mach drag-divergence number (M_{DD}):



- At $M \sim 0.6$, severely reduced L/D.
- Benefit of this design is at $M > 1$, to sweep wings inside Mach cone.

- Wing sweep is beneficial since it increases drag-diverges Mach Number.
- Increasing wing sweep reduces the lift coefficient.

Wing Sweep

- Effect of sweep on wing weight:

$$\boldsymbol{W_{wing}} = CS_W^{0.649} A^{0.5} (t/c)_{root}^{-0.4} (1+\lambda)^{0.1} \boldsymbol{(\cos \Lambda)^{-1}}$$

- *The weight of the wing is proportional to* $\mathbf{1/\cos \Lambda}$.
- Thus, the weight of the wing increases as sweep increases.

Wing Sweep

Why sweep the wing?

Subsonic (usually small):
- Adjust wing aero center relative to center of gravity.
- On flying wing, get moment arm length for control.

Transonic (significant, $30° - 35°$)
- Delay drag rise Mach

Supersonic (large, $45° - 70°$)

Why variable sweep?

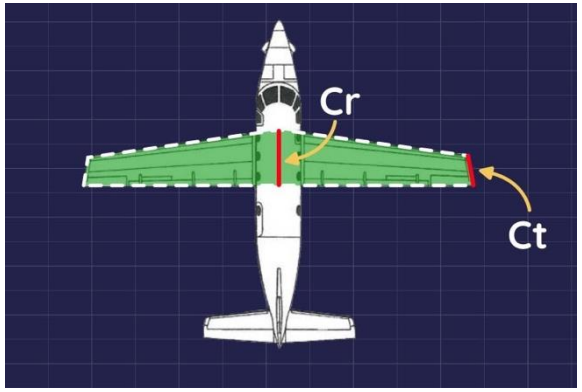
- To get the best of both worlds.
- But: adds weight/complexity, currently unfashionable

Wing sweep increases
wing weight for fixed span



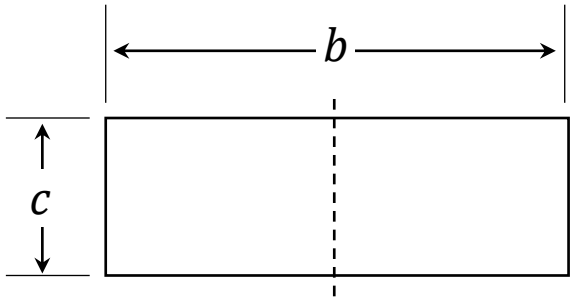
F-14 Tomcat

Taper Ratio



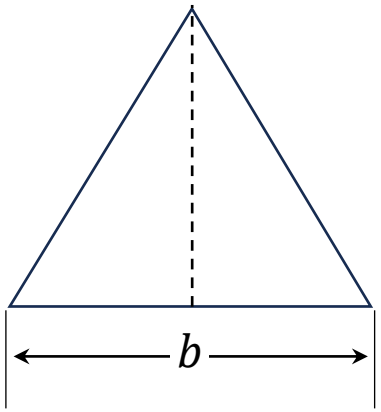
$$\lambda = \frac{c_{tip}}{c_{root}}$$

Rectangular Wing



$\lambda = 1$

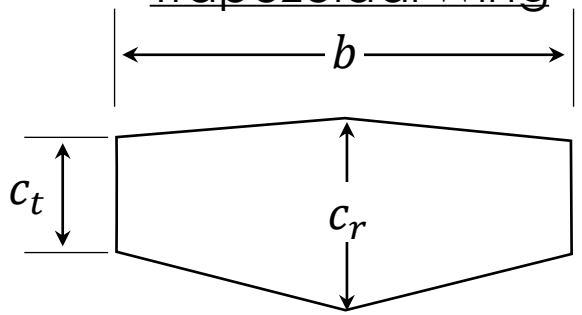
Delta Wing



$c_{tip} = 0$

$\lambda = 0$

Trapezoidal Wing



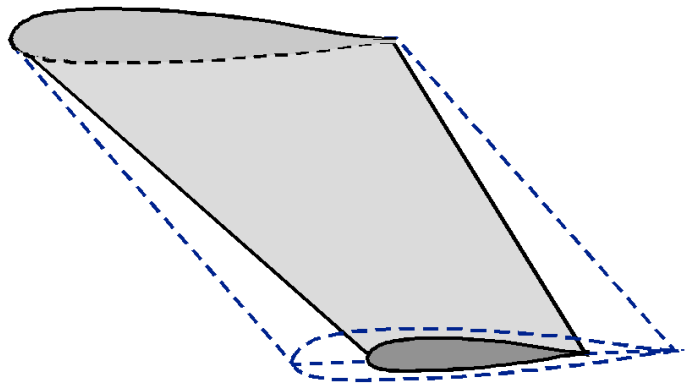
$0 < \lambda < 1$

Taper Ratio

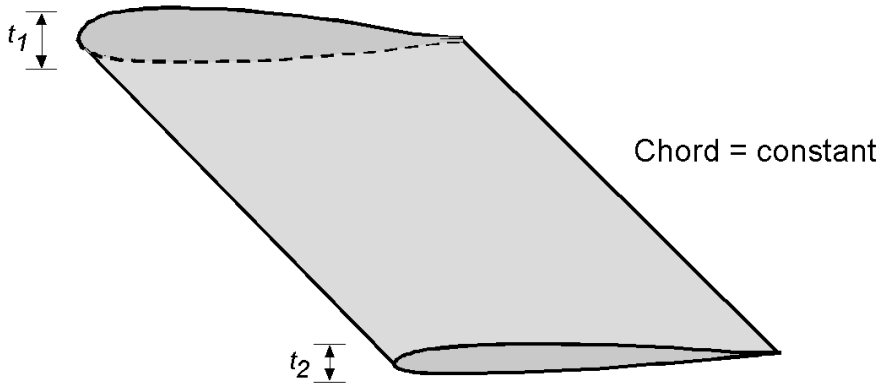
Taper ratio influences:

- Induced drag
- Structural weight
- Ease of fabrication

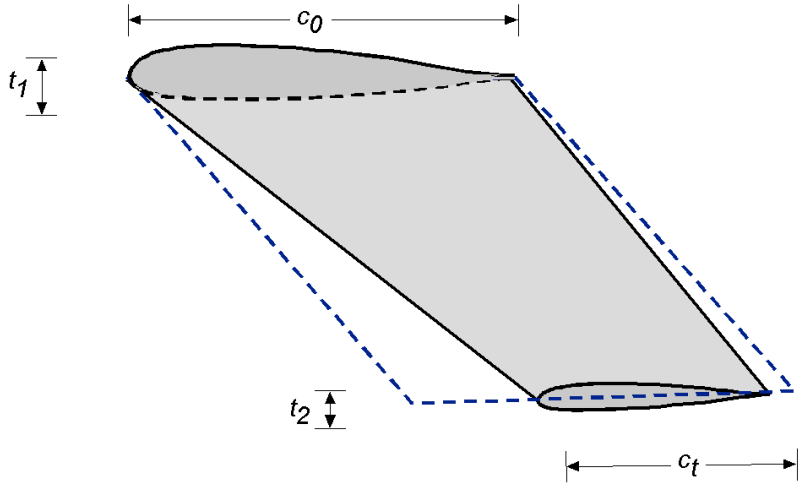
Platform taper



Thickness taper

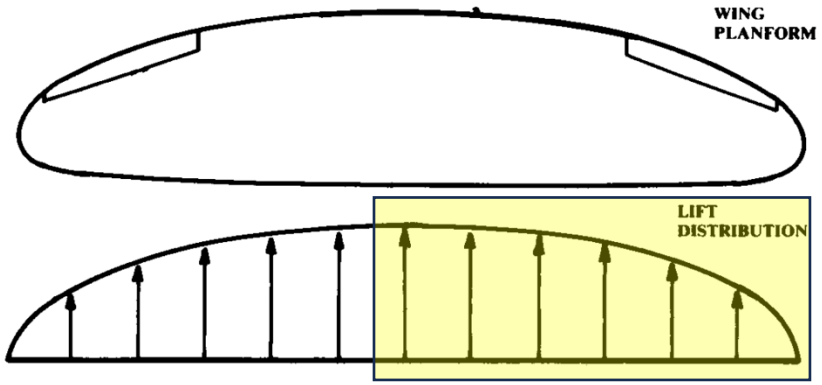


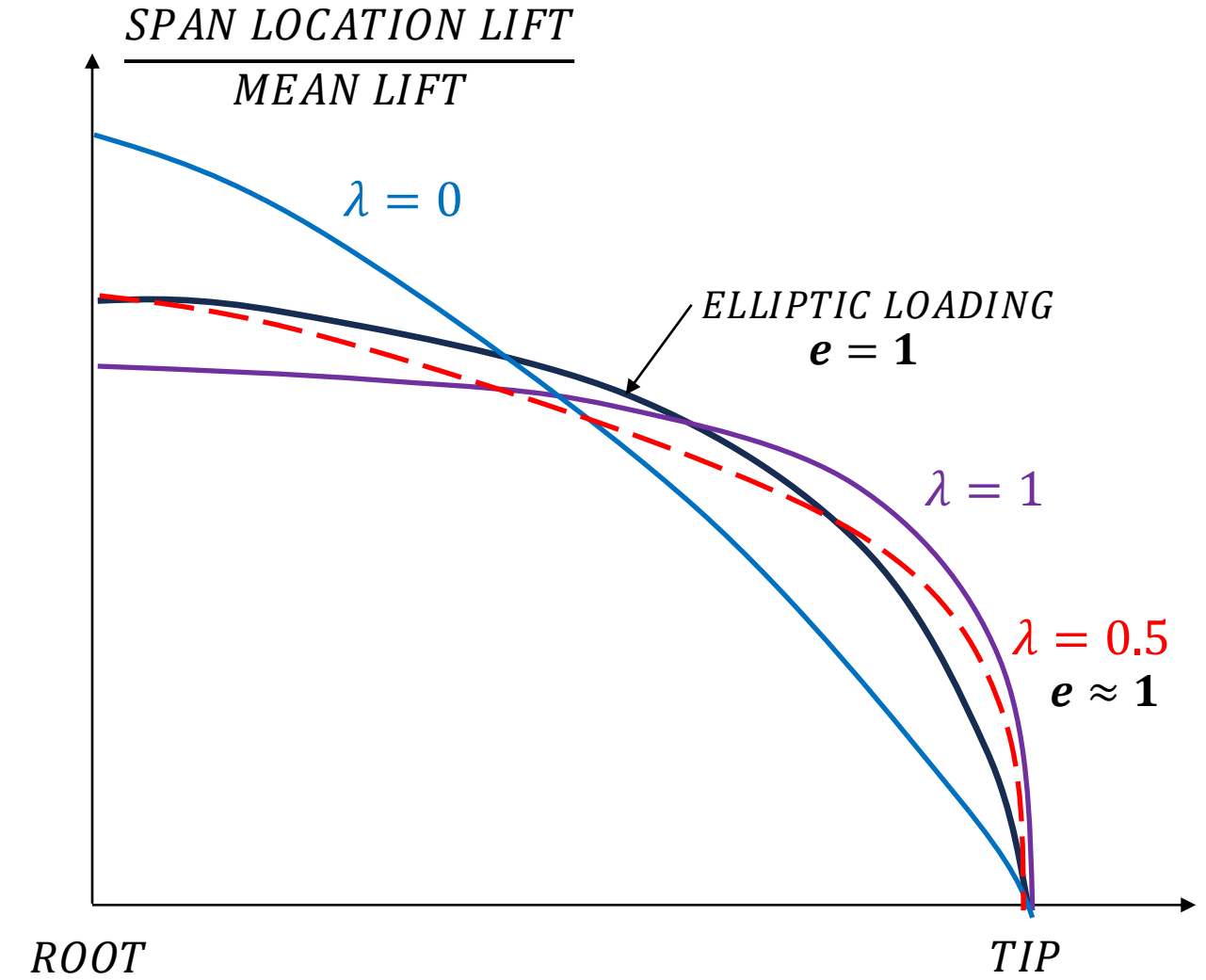
Platform and thickness taper



Taper Ratio

- It is known that an elliptic wing has the lowest induced drag. However this planform shape is difficult to fabricate.



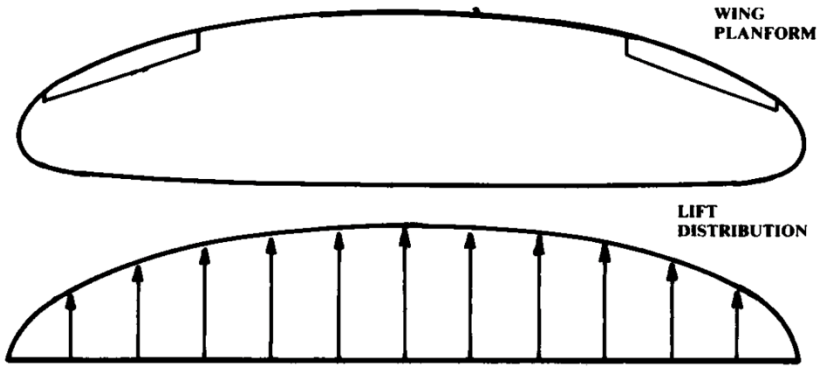


$0 < e < 1$

→ Oswald's Efficiency Factor

Taper Ratio

- It is known that an elliptic wing has the lowest induced drag. However this planform shape is difficult to fabricate.



- A rectangular wing is easy to fabricate but has about 7% higher C_{D_i} as compared to the elliptic wing, it is also heavier structurally.

- Necessary measures for avoiding tip-stalling.
  - An unswept wing, with λ between 0.2 to 0.3.
  - A swept wing, with λ between 0.4 to 0.5.

- Further in a tapered wing, the span loading is concentrated in the inboard portions of the wing and the aerofoil at the root is thicker than near the tip. These factors help in reducing the wing weight.

Mean Aerodynamic Chord

Mean Aerodynamic Chord (MAC)

- The mean aerodynamic chord (MAC) is defined as the chord of an equivalent rectangular wing that would experience aerodynamic forces and moments identical to those of the actual wing. The MAC is defined as:

$$\mathrm{MAC} = \bar{c} = \frac{2\int_0^{b/2} c^2\,dy}{S_w}$$

Half a span

Reference Wing Area

